

ASSEMBLE AND DISASSEMBLE

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

=>

Component	Short Description
Motherboard	The main circuit board that connects and allows communication between all components.
CPU (Processor)	The brain of the computer that executes instructions and performs calculations.
CPU Cooler/Fan	Keeps the CPU cool by dissipating heat generated during operation.
RAM (Memory)	Temporary storage that holds data and programs currently in use.
GPU (Graphics Card)	Processes graphics, videos, and gaming visuals.
HDD (Hard Disk Drive)	Stores operating system, applications, and files using magnetic disks.
SSD (Solid State Drive)	Faster storage device that stores data on flash memory chips.
Power Supply Unit (PSU)	Converts AC power from the wall into DC power for computer components.
Case Fans	Improve airflow inside the cabinet and help keep components cool.
Optical Drive (Optional)	Reads and writes CDs, DVDs, or Blu-ray discs.
PCIe Expansion	Additional cards such as Wi-Fi, sound, or capture cards that add

Cards	extra features.
Cables and Connectors	Connect power and data between different components inside the PC.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

=>Desktop PC Disassembly Documentation

Step 1: Unplug Power Cable

Diagram/Photo:

[Power Cable Connected]
↓
[Power Cable Removed]

Caption:
First, I shut down the computer and unplugged the power cable from the power supply to ensure safety before opening the cabinet.

Step 2: Remove the Side Panel

Diagram/Photo:

+-----+
| PC Cabinet |
+-----+
↓
[Side Panel Removed]

Caption:
I removed the screws securing the side panel and carefully slid the panel off to access the internal components.

Step 3: Identify Internal Components

Diagram/Photo:

```
+-----+
| Motherboard      |
| CPU & Cooler     |
| RAM              |
| GPU              |
| HDD / SSD        |
| PSU              |
+-----+
```

Caption:

After opening the cabinet, I identified the motherboard, CPU cooler, RAM, storage drives, graphics card, and power supply.

Step 4: Disconnect Power and Data Cables

Diagram/Photo:

```
PSU -----> Motherboard
PSU -----> HDD/SSD
SATA -----> HDD/SSD
```

Caption:

I carefully disconnected all power and data cables from the motherboard and storage devices.

Step 5: Remove RAM Modules

Diagram/Photo:

```
RAM Slot
[ RAM ]
  ↓
[Removed]
```

Caption:

I pressed the retention clips on both sides of the RAM slot and gently removed the RAM module.

Step 6: Remove HDD/SSD

Diagram/Photo:

```
+-----+
| HDD/SSD |
+-----+
  ↓
[Removed]
```

Caption:

I unscrewed the storage drive from its mounting bracket and disconnected all cables before removing it.

Step 7: Remove Graphics Card (GPU)

Diagram/Photo:

```
PCIe Slot  
[ GPU ]  
  ↓  
[Removed]
```

Caption:

I removed the retaining screw, released the PCIe slot lock, and carefully pulled out the graphics card.

Step 8: Remove Motherboard

Diagram/Photo:

```
+-----+  
| Motherboard |  
+-----+  
  ↓  
[Removed]
```

Caption:

After disconnecting all cables and removing mounting screws, I carefully lifted the motherboard out of the cabinet.

3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

=>

Desktop PC Reassembly Documentation

Step 1: Install the Motherboard

Explanation:

I carefully placed the motherboard inside the cabinet and secured it with screws. This step is done first because all other components connect to the motherboard.

Step 2: Install the CPU and CPU Cooler

Explanation:

I installed the CPU into the motherboard socket and attached the CPU cooler. The cooler is necessary to prevent the processor from overheating during operation.

Step 3: Install RAM Modules

Explanation:

I inserted the RAM modules into the correct memory slots until the locking clips clicked into place. RAM must be installed before other large components block access to the slots.

Step 4: Install HDD/SSD Storage Drives

Explanation:

I mounted the HDD/SSD in the drive bay and connected the SATA data and power cables. Storage devices are needed to store the operating system and files.

Step 5: Install the Graphics Card (GPU)

Explanation:

I inserted the GPU into the PCIe slot and secured it with screws. The graphics card is installed after the motherboard and RAM because it is a large component that can obstruct access.

Step 6: Connect Power Supply Cables

Explanation:

I connected the 24-pin motherboard power cable, CPU power cable, GPU power cable, and storage power cables. These connections provide power to all components.

Step 7: Connect Front Panel and Data Cables

Explanation:

I connected the power switch, reset switch, USB, audio, and SATA data cables. These connections allow the PC's buttons and ports to function correctly.

Step 8: Check All Connections

Explanation:

I verified that all cables, RAM modules, storage drives, and expansion cards were properly seated. This helps prevent startup or hardware detection issues.

Step 9: Replace the Side Panel

Explanation:

I reattached the side panel and tightened the screws. This protects the internal components and maintains proper airflow.

Step 10: Power On and Test

Explanation:

I connected the power cable and turned on the PC. The system booted successfully, confirming that all components were installed correctly.

4. Assemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

=>Desktop PC Assembly Documentation

Step 1: Prepare the Cabinet

What I Did:

Opened the PC cabinet and checked that all mounting screws and standoffs were in place.

Why the Order Matters:

The cabinet must be ready before installing any components.

Step 2: Install the Motherboard

What I Did:

Placed the motherboard inside the cabinet and secured it with screws.

Why the Order Matters:

The motherboard is the main component to which all other hardware will connect.

Step 3: Install the CPU

What I Did:

Carefully aligned the CPU with the socket and locked it into place.

Why the Order Matters:

The CPU should be installed before the cooler and other components to ensure easy access.

Step 4: Install the CPU Cooler

What I Did:

Attached the heatsink and fan to the CPU and connected the fan cable to the CPU_FAN header.

Why the Order Matters:

The cooler protects the CPU from overheating and is easier to install before other components are added.

Step 5: Install RAM

What I Did:

Inserted the RAM modules into the memory slots until the clips locked.

Why the Order Matters:

Installing RAM early provides easy access to the slots before larger components are installed.

Step 6: Install HDD/SSD

What I Did:

Mounted the HDD/SSD in the drive bay and connected the SATA data and power cables.

Why the Order Matters:

Storage devices should be installed before cable management becomes crowded.

Step 7: Install the GPU

What I Did:

Inserted the graphics card into the PCIe slot and secured it with screws.

Why the Order Matters:

The GPU is large and can block access to other components if installed earlier.

Step 8: Install the Power Supply (PSU)

What I Did:

Mounted the PSU and connected power cables to the motherboard, CPU, GPU, and storage devices.

Why the Order Matters:

Power connections are made after all major hardware is installed.

Step 9: Connect Front Panel Cables

What I Did:

Connected the power switch, reset switch, USB, and audio cables to the motherboard.

Why the Order Matters:

These small cables are easier to connect after the main components are secured.

Step 10: Final Inspection and Power On

What I Did:

Checked all connections, closed the side panel, connected external cables, and powered on the PC.

Why the Order Matters:

A final inspection helps prevent errors and ensures the system boots successfully.
